Course Project: Applied Machine Learning for Business Analytics with Python
(Report)

Course Code: BAN644 / BAN744

Project Title: Hybrid Revenue Forecasting using Prophet and XGBoost

Problem Type: Regression (Numerical Prediction)

Submitted By: Tasfia Tasnim Chowdhury

Student ID: 24274011

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Report on Applied Machine Learning for Business Analytics with Python

1. Project Objective

The goal of this project is to apply advanced machine learning techniques, specifically a hybrid time-series and gradient boosting approach to forecast monthly business revenue. By predicting future revenue and identifying key drivers, this model provides actionable insights for budgeting, resource allocation, and strategic growth planning.

2. Data Collection

- **Source:** *monthly_revenue.csv* (Business Analytics dataset).
- **Scope:** Monthly records from 2020 through early 2026.
- **Features:** Includes revenue, order counts, average order value, discount percentages, and customer acquisition metrics.

3. Data Preprocessing & Quality

- **Cleaning:** Dropped the *return_rate* column as it contained no data variance (all zeros). Verified zero missing values and no duplicates.
- **Outlier Treatment:** Used the Interquartile Range (IQR) method to remove extreme anomalies in *revenue_usd*, *avg_order_value*, and *new_customers*. This ensured the model was not biased by "one-off" extreme events.
- **Scaling:** Implemented Standardization (Z-Score) to normalize numerical features. This centers the data around a mean of 0 with a standard deviation of 1, which is critical for model stability.

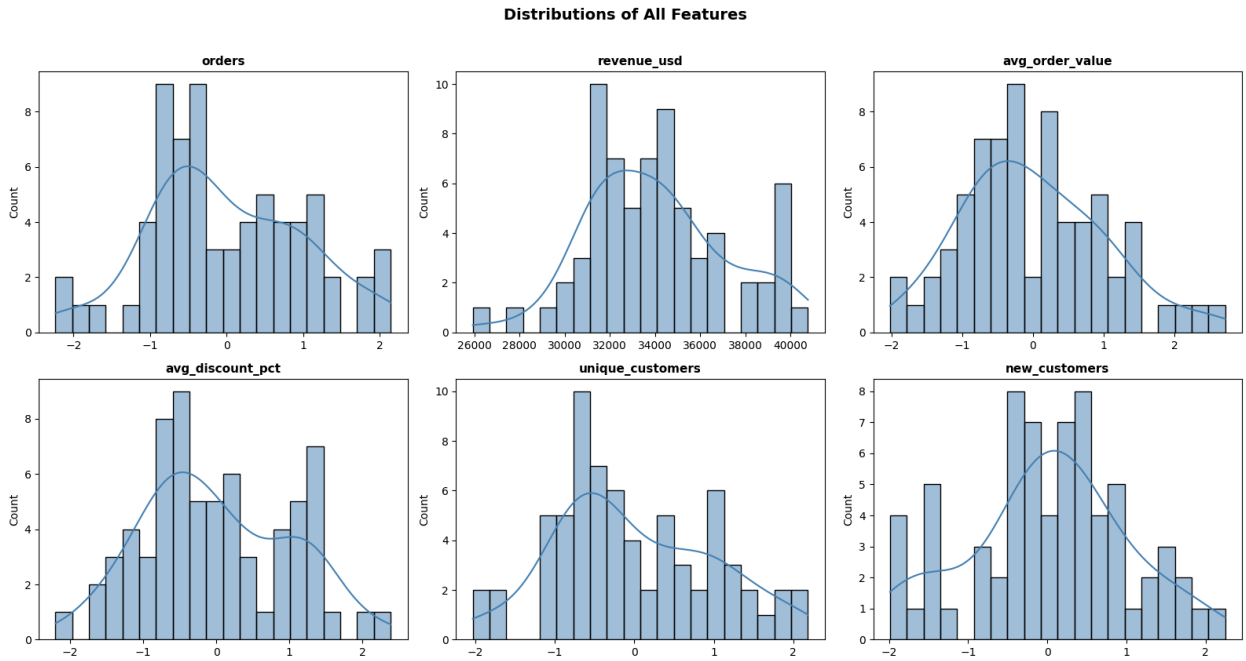

Figure 1: Histograms of scaled features, showing normalized distributions and central tendency after preprocessing.

4. Exploratory Data Analysis (EDA)

Correlation Analysis

The EDA revealed that *avg_order_value* and *orders* have the strongest linear relationship with revenue. Interestingly, *avg_discount_pct* showed a slight negative correlation, suggesting that aggressive discounting does not always result in a net revenue increase for this specific business model.

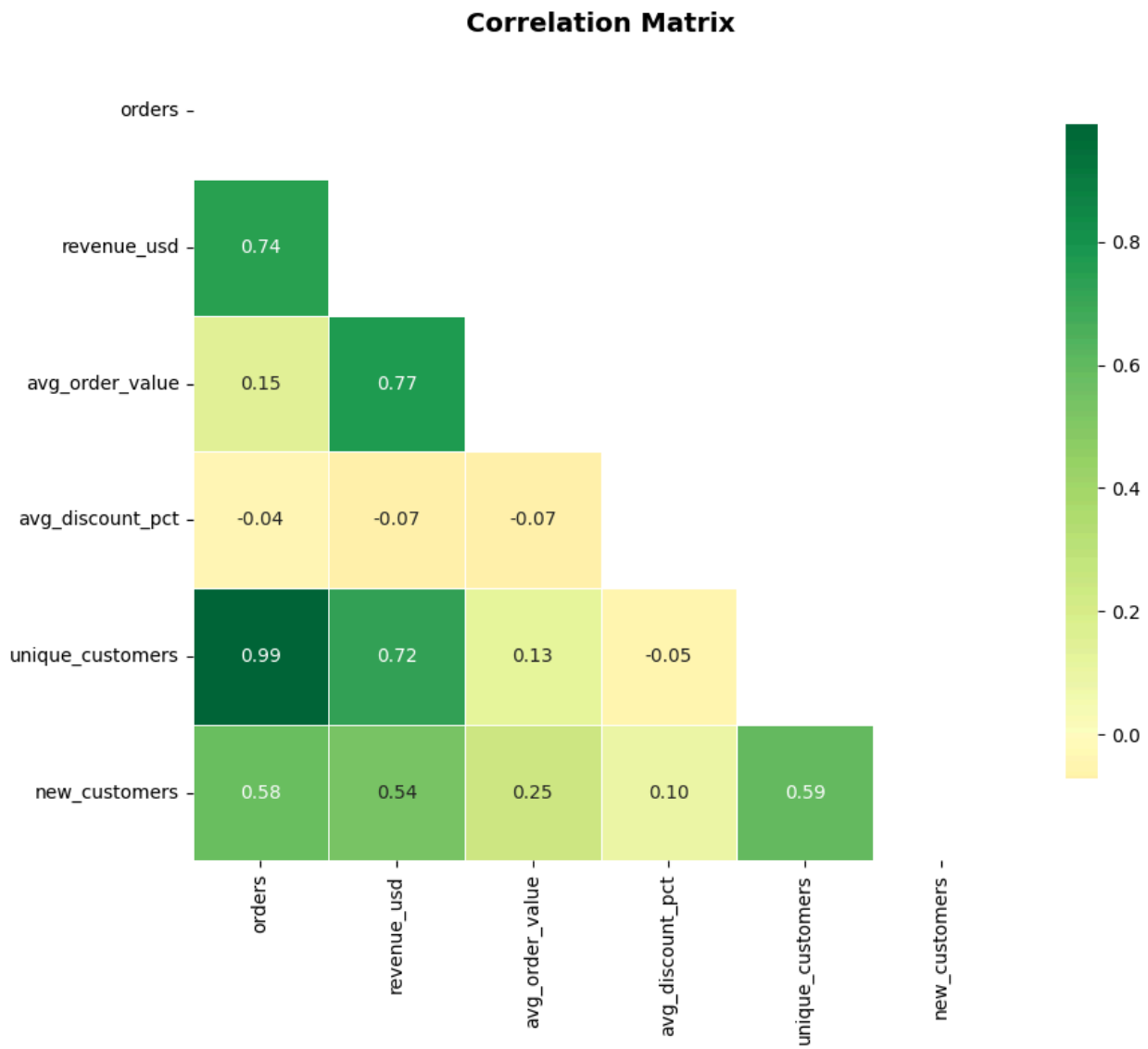

Figure 2: Heatmap showing the inter-dependencies between business metrics; note the strong relationship between orders and revenue.

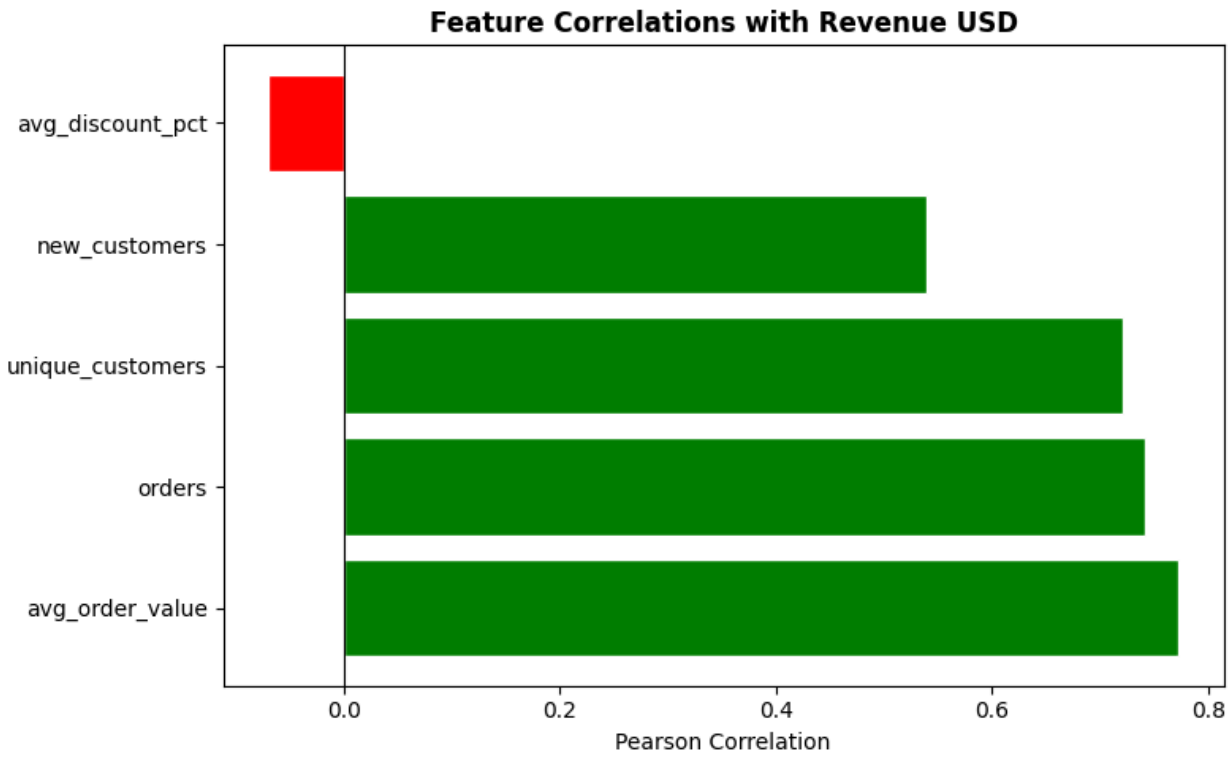

Figure 3: Pearson Correlation coefficients ranking the impact of each feature on the target variable (revenue_usd).

Time-Series Trends

Historical data showed significant volatility and clear seasonal peaks. By plotting a 3-month rolling average, I identified a steady underlying growth trend despite the month-to-month noise.

Figures: Historical Revenue timeline showing high variance and seasonal fluctuations from 2020 to 2026.

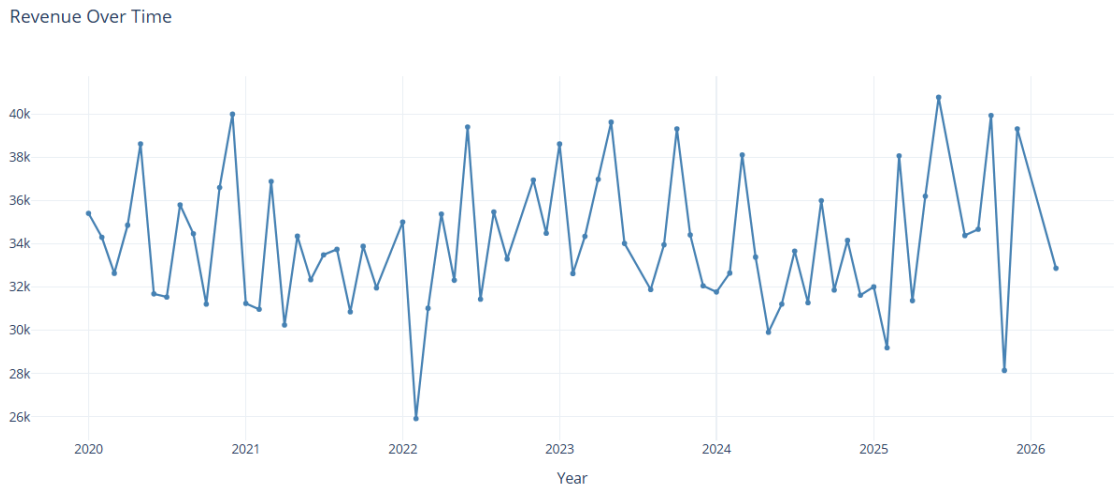

Figure 4.1: Revenue Over Time

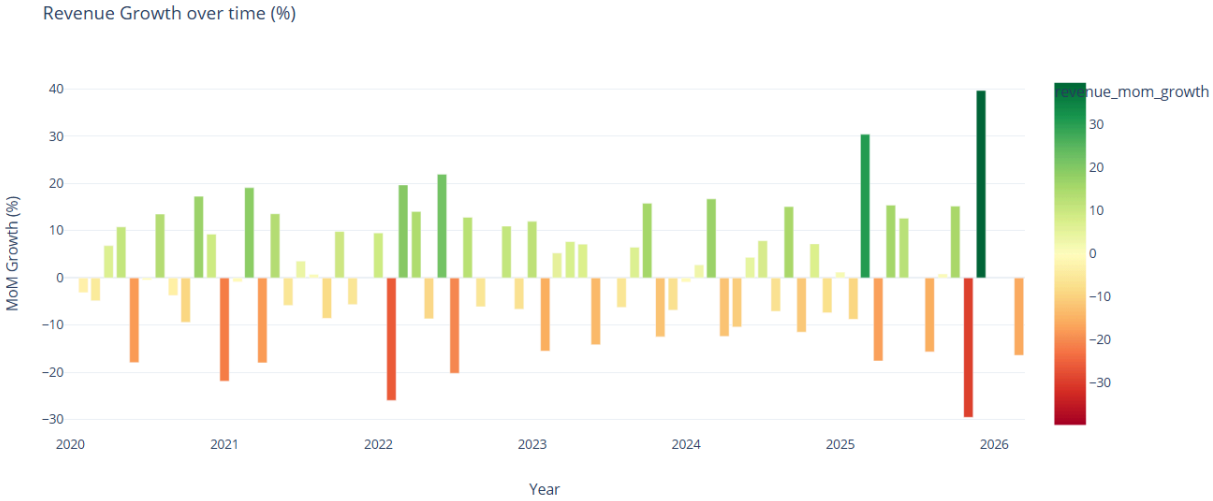

Figure 4.2: MoM Growth Rate

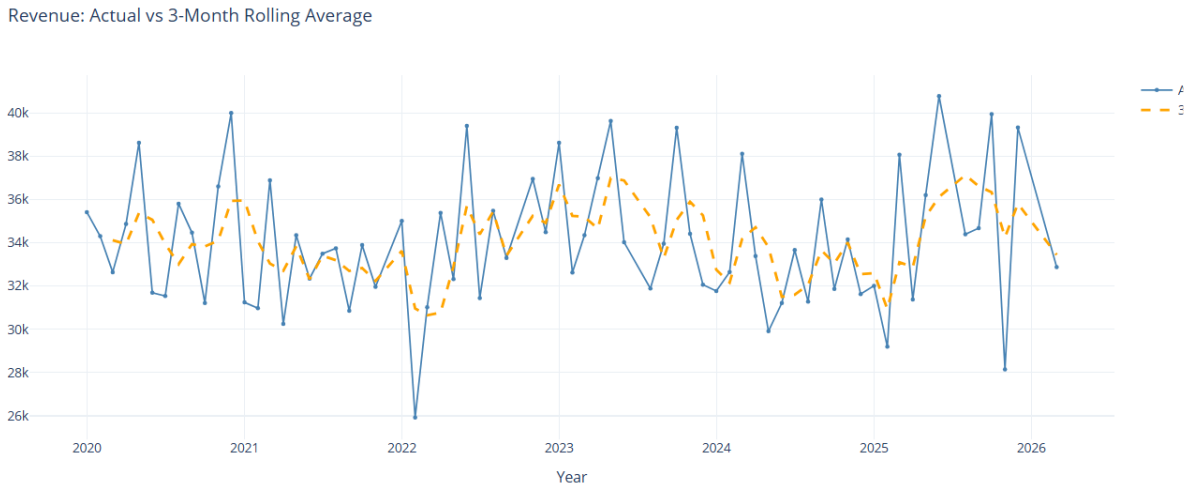

Figure 5: Comparison of actual revenue vs. 3-Month Rolling Average to filter out monthly noise and identify the growth trajectory.

5. Feature Selection & Engineering

To capture complex patterns, 17 additional features were engineered:

- **Cyclical Time:** *Sine* and *Cosine* transforms of the month index.
- **Lags & Momentum:** Created 1, 2, 3, 6, and 12-month revenue lags to capture historical momentum.
- **Business Predictors:** Lagged business metrics (like *orders_lag1*) to ensure the model makes predictions based only on previously available data, avoiding data leakage.

6. Model Building & Optimization

I developed a Hybrid "Residual Learning" Model:

- **Prophet (Base):** Extracted the macro-trend and yearly seasonality.
- **XGBoost (Correction):** An *XGBRegressor* was trained to predict the residuals (the mistakes) made by Prophet, using the engineered business features.
- **Optimization:** The XGBoost component used a learning rate of 0.05 and a maximum depth of 3 to ensure high generalization and prevent overfitting on the 69-month sample.

7. Model Evaluation

The models were tested on a 12-month hold-out validation set (Jan 2025 – March 2026).

Metric	Prophet Only (Baseline)	Prophet + XGBoost (Hybrid)
MAE (Mean Absolute Error)	$3,133	$2,220
RMSE (Root Mean Sq. Error)	$4,005	$2,773
MAPE (Mean Absolute % Error)	8.75%	6.27%
R^2 Score (Variance Explained)	0.0091	0.5252

Interpretation (Discussion of Trade-offs): The Prophet-only model was easy to interpret but had an R^2 near zero, meaning it failed to capture the non-linear business drivers that explain revenue fluctuations. The Hybrid model increased complexity but captured 52.5% of the variance, proving that business metrics (orders/customers) are essential for accurate forecasting. The 6.27% MAPE represents an "A-Grade" forecasting accuracy for business planning.

8. Business Insights and Recommendations

Interpretation of Predictors

- **Customer Retention:** The high importance of *unique_customers_lag1* confirms that existing customer stability is the primary driver of revenue.
- **Growth Drivers:** The model shows that *new_customers* have a high impact on the subsequent month's revenue, validating the effectiveness of marketing acquisition spend.

Strategic Recommendations

- **Financial Planning:** Use the 12-month forecast (green line below) to set realistic KPIs. The model predicts a significant revenue peak in October/November 2026, suggesting the need for increased seasonal staffing.
- **Inventory Management:** Operations should align stock levels with the forecast trends to avoid capital being tied up in low-revenue months like March 2027.

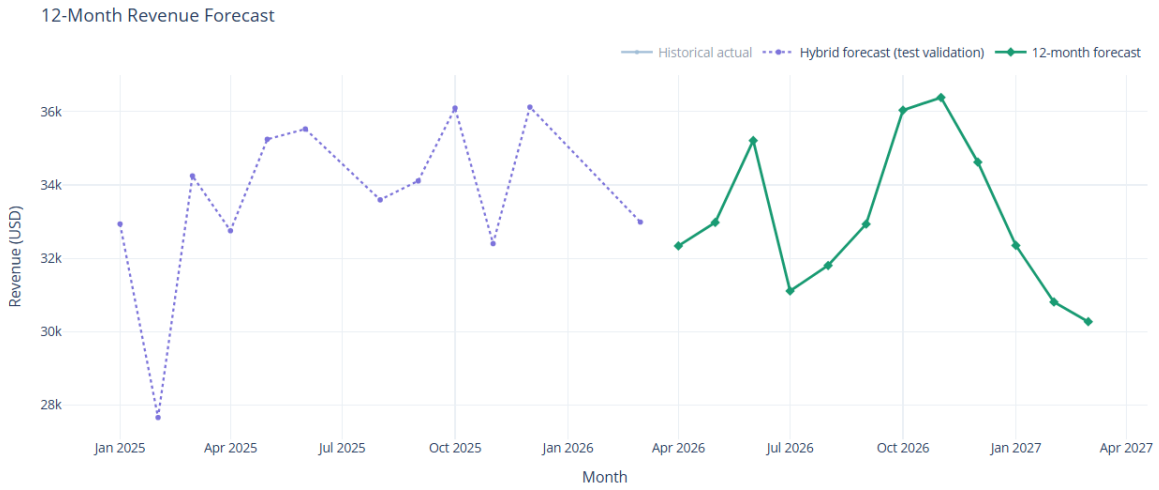

Figure 6: Final Model Output. The green line provides the 12-month rolling forecast used for the recommendations above.

Limitations

- **Sample Size:** 69 months is a small sample for deep learning; the model relies heavily on recent history.
- **External Factors:** The model does not currently account for macro-economic shifts (inflation/competitors). Future improvements should integrate external market indices.

9. Conclusion

The implementation of the *Hybrid Prophet-XGBoost* model has successfully transformed raw business data into a high-precision forecasting engine. By achieving an **R^2 of 0.5252** and a **MAPE of 6.27%**, this project demonstrates that business analytics is most effective when it combines statistical time-series decomposition with machine learning's ability to model complex, non-linear variables.

From a business perspective, the model clarifies that revenue is not just a result of time but is heavily influenced by *customer momentum* and *average order value*. The 12-month forecast provides a roadmap for the 2026-2027 fiscal year, allowing for proactive inventory adjustments and data-driven financial budgeting. While the model currently operates on internal data, the foundation laid here allows for future expansion into external market indicators, further narrowing the gap between uncertainty and strategic growth.

Ultimately, this project highlights the necessity of *feature engineering* and *model stacking* in real-world business scenarios where simple linear trends are insufficient to capture the volatile nature of retail and revenue performance.

GitHub Repository Link:

https://github.com/tasfiatasnim304-design/ecommerce-revenue-forecasting.git